

RVHS H2 Maths 2020 Prelim Paper 2 (Solutions)

Q1

(i) $\triangle ACB$ is a right-angle triangle. $\therefore BC = \sqrt{13^2 - 12^2} = 5$ Hence $\tan \angle BAC = \frac{BC}{AC} = \frac{5}{12}$ (shown)	
(ii) $CX = 12 \tan(\angle BAC - \alpha)$ $= \frac{12(\tan \angle BAC - \tan \alpha)}{1 + \tan \angle BAC \tan \alpha}$ $= \frac{12\left(\frac{5}{12} - \tan \alpha\right)}{1 + \frac{5}{12} \tan \alpha}$ $= \frac{5 - 12 \tan \alpha}{1 + \frac{5}{12} \tan \alpha}$ $= \frac{60 - 144 \tan \alpha}{12 + 5 \tan \alpha}$ (shown)	
(iii) For small α, $\tan \alpha \approx \alpha$ $CX = \frac{60 - 144 \tan \alpha}{12 + 5 \tan \alpha}$ $\approx (60 - 144\alpha)(12 + 5\alpha)^{-1}$ $= (5 - 12\alpha)\left(1 + \frac{5}{12}\alpha\right)^{-1}$ $= (5 - 12\alpha)\left(1 - \frac{5}{12}\alpha + \left(\frac{5}{12}\alpha\right)^2 + \dots\right)$ $\approx 5 - 12\alpha - \frac{25}{12}\alpha + 5\alpha^2 + \frac{125}{144}\alpha^2$ $= 5 - \frac{169}{12}\alpha + \frac{845}{144}\alpha^2$ $p = -\frac{169}{12}, \quad q = \frac{845}{144}$ (shown)	
(iv) For $\alpha = 0.6$, $\tan 0.6 = 0.684 \approx 0.7 \neq 0.6$ Hence, $\tan \alpha \approx \alpha$ is not valid.	

Q2.

(i) $y = ve^{2t}$ $\frac{dy}{dt} = e^{2t} \frac{dv}{dt} + 2ve^{2t} (*)$ $v \frac{dv}{dt} + 2v^2 = te^{-2t}$ $ve^{2t} \frac{dv}{dt} + 2v^2 e^{2t} = t$ $e^{2t} \frac{dv}{dt} + 2ve^{2t} = tv^{-1}$ Replace LHS with (*) $\frac{dy}{dt} = tv^{-1} = t \cdot \frac{e^{2t}}{y}$ $y \frac{dy}{dt} = t e^{2t} \text{ (shown)}$	
(ii) Using separable, $\int y \, dy = \int te^{2t} \, dt$ $\frac{y^2}{2} = \frac{te^{2t}}{2} - \int \frac{e^{2t}}{2} \, dt$ $\frac{y^2}{2} = \frac{te^{2t}}{2} - \frac{e^{2t}}{4} + c$ $y^2 = te^{2t} - \frac{e^{2t}}{2} + D, \text{ where } 2c = D$ $v^2 e^{4t} = te^{2t} - \frac{e^{2t}}{2} + D$ From $v \frac{dv}{dt} + 2v^2 = te^{-2t}, t = 0, \frac{dv}{dt} = -8,$ $-8v + 2v^2 = 0$ $-8 + 2v = 0 \text{ since } v \neq 0$ $v = 4$ Substituting the values of v and t, $v^2 e^{4t} = te^{2t} - \frac{e^{2t}}{2} + D$ $16 = -\frac{1}{2} + D$ $D = 16.5$ Therefore, $v^2 e^{4t} = te^{2t} - \frac{e^{2t}}{2} + 16.5$ $v^2 = te^{-2t} - \frac{e^{-2t}}{2} + (16.5)e^{-4t}$	

Q3.

(i) Let $y = \frac{2x+p}{x-2}, x \in \mathbb{R}, x < 2,$ $y(x-2) = 2x+p$ $yx-2x = 2y+p$ $x(y-2) = 2y+p$ $x = \frac{2y+p}{y-2}$ $\therefore f^{-1}(x) = \frac{2x+p}{x-2}, x < 2$ Since $f = f^{-1}$, f is an involuting function. (SHOWN)	
(ii) From (i), $f(x) = f^{-1}(x)$ Composing function on both sides, $ff(x) = ff^{-1}(x)$ $f^2(x) = x \text{-----} (*)$ $f^3(x) = f(x) = \frac{2x+p}{x-2}$ Rewriting (*), $f^3(x) = f^{-1}(x)$ since f involuting $f^4(x) = f^2(x) = x$ Hence, $f^{2021}(x) = f(x) = \frac{2x+p}{x-2}$	
(iii) Now $f(x) = \frac{2x+2}{x-2}, x < 2$ $g(x) = \ln(4-x), x < 3$ As $R_g = (0, \infty) \not\subseteq (-\infty, 2) = D_f$, fg does not exist. Since $R_f = (-\infty, 2) \subseteq (-\infty, 3) = D_g$, gf exists. $gf(x) = g\left(\frac{2x+2}{x-2}\right)$ $= \ln\left(4 - \frac{2x+2}{x-2}\right)$ $= \ln\left(\frac{2x-10}{x-2}\right)$ $gf : x \mapsto \ln\left(\frac{2x-10}{x-2}\right), x \in \mathbb{R}, x < 2$	

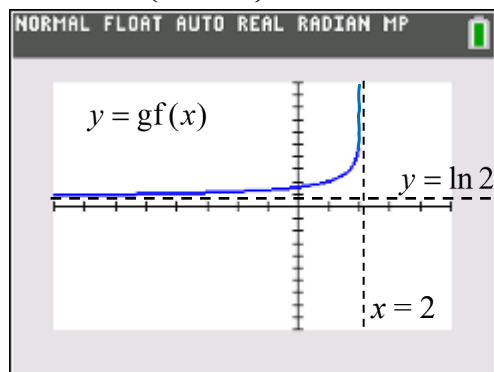
(iv)

Using mapping method ,

$$D_f = (-\infty, 2) \xrightarrow{f} R_f = (-\infty, 2) \xrightarrow{g} R_{gf} = (\ln 2, \infty)$$

Therefore range of gf is $(\ln 2, \infty)$.**Alternatively,**

$$gf(x) = \ln\left(\frac{2x-10}{x-2}\right), x < 2$$



$$\begin{aligned} gf(x) &= \ln\left(\frac{2x-10}{x-2}\right) \\ &= \ln\left(2 - \frac{6}{x-2}\right) \end{aligned}$$

As $x \rightarrow 2$, $gf(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $gf(x) \rightarrow \ln 2$ Therefore range of gf is $(\ln 2, \infty)$.**Q4.**

(i)

$$l: r = \begin{pmatrix} 9 \\ -5 \\ 2 \end{pmatrix} + \beta \begin{pmatrix} a \\ b \\ 1 \end{pmatrix}, \beta \in \mathbb{R}$$

Since line is parallel to plane, the direction vector of line is perpendicular to the normal of the plane;

$$\begin{pmatrix} a \\ b \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -4 \\ -1 \end{pmatrix} = 0$$

$$3a - 4b = 1$$

(ii)(a)

$$\begin{pmatrix} 9+2\beta \\ -5-2\beta \\ 2+\beta \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -4 \\ -1 \end{pmatrix} = -7$$

$$27 + 6\beta + 20 + 8\beta - 2 - \beta = -7$$

$$13\beta = -52$$

$$\beta = -4$$

The position vector of the point of intersection is

$$\begin{pmatrix} 9 \\ -5 \\ 2 \end{pmatrix} + (-4) \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$$

(ii)(b)

Construct a line parallel to the normal of the plane and passing

through point A: $r = \begin{pmatrix} 9 \\ -5 \\ 2 \end{pmatrix} + \alpha \begin{pmatrix} 3 \\ -4 \\ -1 \end{pmatrix}, \alpha \in \mathbb{R}.$

Subs. equation of line into that of plane:

$$\begin{pmatrix} 9+3\alpha \\ -5-4\alpha \\ 2-\alpha \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -4 \\ -1 \end{pmatrix} = -7$$

$$27+9\alpha+20+16\alpha-2+\alpha=-7$$

$$\alpha = -2$$

The position vector of the foot of perpendicular is

$$\begin{pmatrix} 9 \\ -5 \\ 2 \end{pmatrix} + (-2) \begin{pmatrix} 3 \\ -4 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix}$$

$$\text{Distance} = \left| \begin{pmatrix} 9-3 \\ -5-3 \\ 2-4 \end{pmatrix} \right| = \sqrt{6^2 + (-8)^2 + (-2)^2} = \sqrt{104} \text{ units}$$

Distance of A to the point of intersection =

$$\left| \begin{pmatrix} 9-1 \\ -5-3 \\ 2+2 \end{pmatrix} \right| = \sqrt{8^2 + (-8)^2 + (4)^2} = \sqrt{144} = 12$$

$$\sin \theta = \frac{\sqrt{104}}{12}$$

$$\theta = 58.2^\circ$$

Q5.

(a) Case (1): only 1 P used: $_ _ P$ No. of required ways = $\binom{3}{2} \times 3! = 18$ Case (2) : 2 P's used: $_ P P$ No. of required ways = $\binom{3}{1} \times \frac{3!}{2!} = 9$ Total required ways = $18 + 9 = 27$	
(b)(i) Required Number of ways = $(9 - 1)! = 40320$	
(b)(ii) There is only 1 way to arrange the female teachers in order of height. Next, position a male teacher in each of the slots between 2 female teachers: $F _ F _ F _ F _ F$ No. of ways = $4! = 24$	
(b)(iii) No. of ways the male teachers are arranged among themselves = $4! = 24$ No. of ways the female teachers are arranged among themselves = $5! = 120$ There are 2 ways to arrange the two groups of male and female teachers. Total no. of ways = $2 \times 24 \times 120 = 5760$	

Q6.

(i)(a) $\frac{5+6}{36} = \frac{11}{36}$	
(i)(b) $\frac{5+8+7}{36} = \frac{5}{9}$	
(ii)(a) $P(\text{both are Aces} \mid \text{neither of the cards is Spade})$ $= \frac{P(\text{both Aces \& not Spade})}{P(\text{not Spade})}$ $= \frac{\binom{5}{2}}{\binom{30}{2}} = \frac{2}{87}$	
(ii)(b) $P(\text{'Queen not Diamond' AND 'Diamond not Queen'}) + P(\text{'Queen Diamond' AND 'Not Queen Not Diamond'})$ $= \frac{\binom{9}{1}\binom{6}{1}}{\binom{36}{2}} + \frac{\binom{4}{1}\binom{17}{1}}{\binom{36}{2}} = \frac{61}{315}$	

Q7.

(i) Given: $\sum(x-12) = 39$, $\sum(x-12)^2 = 2716.88$ Unbiased estimate of the population mean, $\bar{x} = \frac{39}{100} + 12 = 12.39$ Unbiased estimate of the population variance, $s^2 = \frac{1}{99} \left(\sum(x-12)^2 - \frac{(\sum(x-12))^2}{100} \right)$ $= 27.2896 = 27.3 \text{ (to 3 s.f.)}$	
(ii) The researcher should carry out a one-tail test to test whether the data of incubation period for Mar <u>has indeed increased</u> as a few data points of 20 days raised an alarm. Let μ be the population mean incubation period for patients in Atopia. To test $H_0: \mu = 10.8$ against $H_1: \mu > 10.8$ at 1% significance level	
(iii) Test statistic: Under H_0, $\bar{X} \sim N\left(10.8, \frac{s^2}{100}\right)$ approximately by CLT since n is large $\Rightarrow Z = \frac{\bar{X} - 10.8}{s / \sqrt{100}} \sim N(0, 1) \text{ approximately}$ $p\text{-value} = 0.00117 < 0.01,$ we reject H_0. There is sufficient evidence at 1% significance level that the incubation period has increased.	
(iv) To test $H_0: \mu = 10.8$ against $H_1: \mu \neq 10.8$ 2-tail test at 1% significance level Under H_0, $Z = \frac{\bar{X} - 10.8}{s / \sqrt{100}} \sim N(0, 1)$ Critical region to reject H_0: By GC, $z_{calc} < -2.5758293$ or $z_{calc} > 2.5758293$	

Since H_0 is rejected,

$$\frac{\bar{x} - 10.8}{s/10} < -2.5758293 \text{ or } \frac{\bar{x} - 10.8}{s/10} > 2.5758293$$

$$\Rightarrow \bar{x} < 9.45 \text{ or } \bar{x} > 12.1 \text{ (to 3sf)}$$

Q8.

(i)

- Assume that the probability p is constant for each worker.
- Assume that the event that any particular worker is sick is independent of the other workers.

1st assumption can be invalid as some workers maybe more likely to fall sick than others.

2nd assumption can be invalid as if one worker is sick, he can spread his sickness to others.

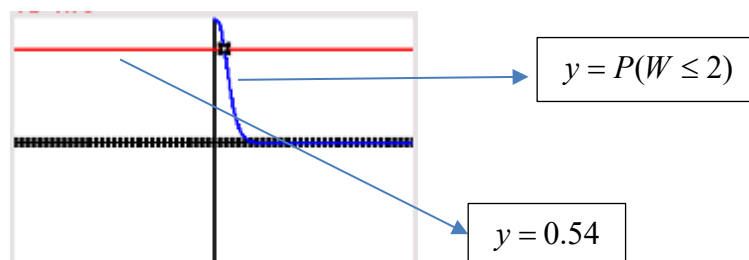
(ii)

$$W \sim B(50, p)$$

$$P(W \leq 2) = 0.54$$

$$\binom{50}{0} p^0 (1-p)^{50} + \binom{50}{1} p^1 (1-p)^{49} + \binom{50}{2} p^2 (1-p)^{48} = 0.54$$

$$(1-p)^{50} + 50p(1-p)^{49} + 1225p^2(1-p)^{48} = 0.54$$



From GC, $p = 0.0500404 = 0.0500$ (to 3 s.f.)

(iii)

Let X be the number of days (out of 60) for which there are at most 2 workers on sick leave.

$$X \sim B(60, 0.54)$$

$$E(X) = 60(0.54) = 32.4$$

(iv)

$$\text{Given } W \sim B(50, 0.0500404)$$

$$\Rightarrow E(W) = 50(0.0500404) = 2.50202 \text{ and}$$

$$\text{Var}(X) = 50(0.0500404)(0.9499596) = 2.37682$$

Since $n = 60$ is large, by Central Limit Theorem,

$$\bar{W} \sim N\left(2.50202, \frac{2.37682}{60}\right) \text{ approximately}$$

$$P(\bar{W} \geq 2) = 0.994$$

Q9.

$$(i) \frac{\binom{4}{1}\binom{3}{3}}{\binom{7}{4}} = \frac{4}{35}$$

(ii)

x	1	2	3	4
$P(X=x)$	$\frac{\binom{4}{1}\binom{3}{3}}{\binom{7}{4}}$ $= \frac{4}{35}$	$\frac{\binom{4}{2}\binom{3}{2}}{\binom{7}{4}}$ $= \frac{18}{35}$	$\frac{\binom{4}{3}\binom{3}{1}}{\binom{7}{4}}$ $= \frac{12}{35}$	$\frac{1}{35}$

(iii)

$$E(X) = 1 \cdot \frac{4}{35} + 2 \cdot \frac{18}{35} + 3 \cdot \frac{12}{35} + 4 \cdot \frac{1}{35} = \frac{16}{7}$$

$$E(X^2) = 1^2 \cdot \frac{4}{35} + 2^2 \cdot \frac{18}{35} + 3^2 \cdot \frac{12}{35} + 4^2 \cdot \frac{1}{35} = \frac{40}{7}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{40}{7} - \left(\frac{16}{7}\right)^2 = \frac{24}{49}$$

(iv)

Let X_1, X_2, X_3 be the correct guesses by 3 listeners respectively.

$$P(X_1 + X_2 + X_3 = 6)$$

$$= P(X_1 = 1) \cdot P(X_2 = 2) \cdot P(X_3 = 3) \times 3! +$$

$$P(X_1 = 2) \cdot P(X_2 = 2) \cdot P(X_3 = 2) +$$

$$P(X_1 = 1) \cdot P(X_2 = 1) \cdot P(X_3 = 4) \times 3$$

$$= \left(\frac{4}{35}\right)\left(\frac{18}{35}\right)\left(\frac{12}{35}\right) \times 6 + \left(\frac{18}{35}\right)^3 + \left(\frac{4}{35}\right)^2 \left(\frac{1}{35}\right) \times 3$$

$$= 0.258$$

One assumption we have made is that the guesses of listeners are independent of each other.

(v)

Required probability

$$= 35 \times \left(\frac{1}{35}\right)^3$$

$$= \frac{1}{1225}$$

Alternatively

$$\frac{\binom{7}{4}}{\binom{7}{4}^3} = \frac{1}{1225}$$

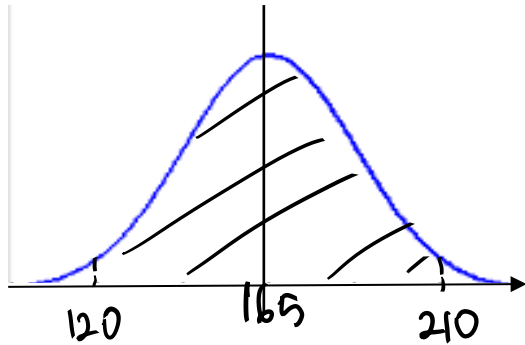
Q10.

(i)

$165 \pm 3(15)$ gives a range of (120,210).

Let P be the mass of phone body in grams.

$$P \sim N(165, 15^2)$$



(Note: 99.7% of the data will lie within 3 standard deviations, thus the shading should take up almost the whole area)

(ii)

$$P_1 - P_2 \sim N(0, 2(15^2)) = N(0, 450)$$

$$P(|P_1 - P_2| < 0.5)$$

$$= P(-0.5 < P_1 - P_2 < 0.5)$$

$$= 0.0188$$

(iii)

Let B be the mass of a battery, i.e., $B = 0.18P$

$$B = 0.18P \sim N(0.18(165), 0.18^2(15^2))$$

$$= N(29.7, 7.29)$$

$$Y = P + B \sim N(165 + 29.7, 15^2 + 7.29)$$

$$= N(194.7, 232.29)$$

P (ideal mass)

$$= P(140 < Y < 170)$$

$$= 0.0524 \text{ (to 3 sf, by GC)}$$

(Note: $\text{Var}(P + B) \neq \text{Var}(1.18P)$)

(iv)

Let X be the random variable for the total mass of the phone body, battery and travel charger

$$X = Y + q, \text{ where } q \text{ is a constant}$$

$$X \sim N(194.7 + q, 232.29), \text{ i.e., } N(194.7 + q, 232.29)$$

$$P(X < 247) = 0.85$$

$$P\left(Z < \frac{247 - (194.7 + q)}{\sqrt{232.29}}\right) = 0.85$$

$$\text{By GC, } \frac{52.3 - q}{\sqrt{232.29}} = 1.03643338$$

$$q = 36.504 = 36.5 \text{ g}$$